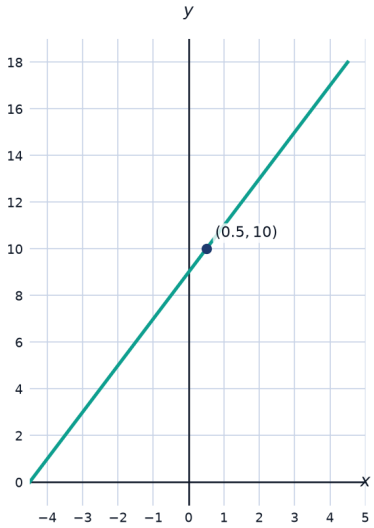
Transformations & Vectors

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

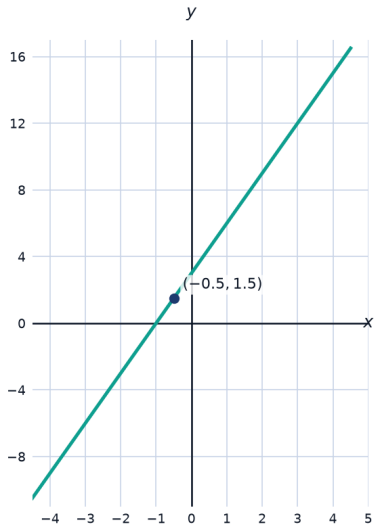
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 2x + 9$ in solving equations with algebra, evaluate $f(5)$.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 3$ in methods for solving two-step equations, evaluate $f(2)$.



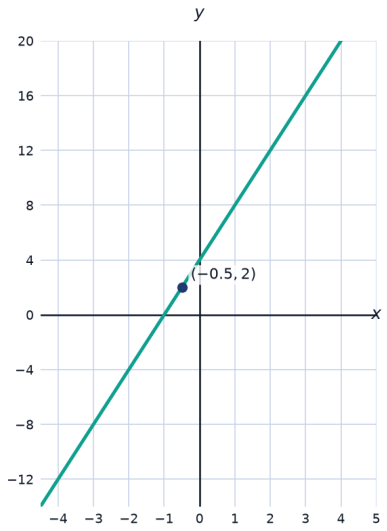
3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for methods for solving two-step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 4$ in two-step equations, evaluate $f(2)$.

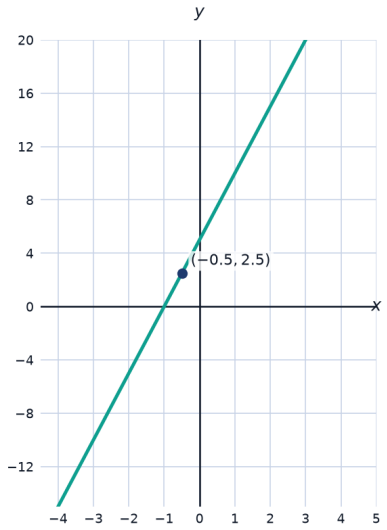


5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for two-step equations consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 5x + 5$ in solving problems with equations, evaluate $f(2)$.



7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for solving problems with equations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

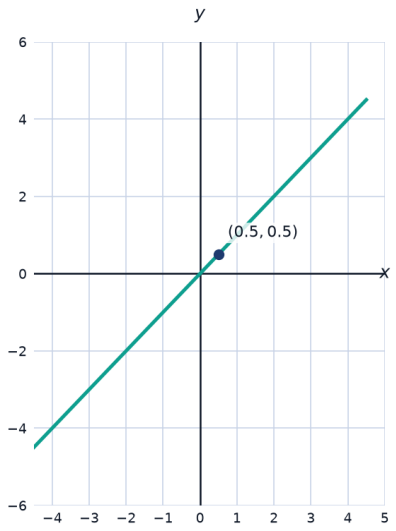
The curve $y = x^2 - 6x + 5$ can be mapped to $y = x^2$ by a sequence of two geometric transformations. Express the equation in vertex form, state the vertex, and describe fully each transformation.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Triangle ΔABC has vertices $A(1, 2)$, $B(3, 2)$, and $C(1, 5)$. The triangle is dilated by a scale factor of $k = 3$ centered at the origin, and then translated by vector $\langle -2, 4 \rangle$. What are the coordinates of vertex C''?

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Point $A(-2, 4)$ is reflected across the line $y = x$, and the resulting point is then translated by the vector $\langle 3, -1 \rangle$. What are the final coordinates?



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x+4}{2x-5}$, and state the domain of f^{-1}.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Given $f(x) = \frac{2}{x-1}$ and $g(x) = \sqrt{x+5}$, find (a) the value of $(f \circ g)(4)$, and (b) the domain of $g(x)$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The curve $y = f(x)$ has a stationary point at $(4, -6)$. State the coordinates of the corresponding stationary point on the curve: (a) $y = f(x + 3) + 2$, (b) $y = 3f(2x)$, (c) $y = -f(-x)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Functions & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 3.01 Coordinate Geometry**

[Transformations & Symmetry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 4.01 Mensuration**

[Graphs & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 5.01 Geometry**

[Vectors in Two Dimensions & Relative Motion · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 6.01 Trigonometry**

A car of mass $1200\ \text{kg}$ ascends a hill of inclination α, where $\sin\alpha = \frac{1}{20}$, against a constant resistance to motion of $500\ \text{N}$. Taking $g = 9.8\ \text{m/s}^2$: (a) find the driving force required to travel at a steady speed of $25\ \text{m/s}$, (b) calculate the power developed by the engine in kilowatts.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 8.01 Probability & Statistics**

Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cup B) = 0.8$. Find $P(A \cap B)$, calculate $P(A|B)$, and determine whether A and B are independent.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 9.01 Functions**

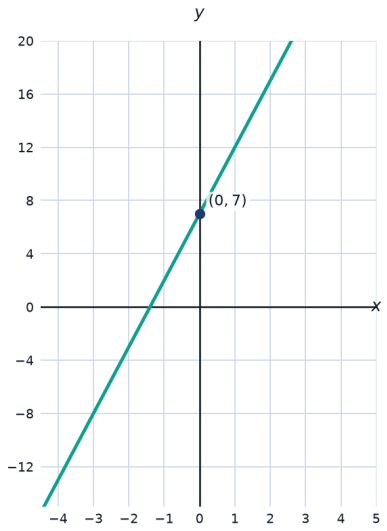
Find the sum of all integers between 10 and 100 (inclusive) that are divisible by 4.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 7$ in introduction to equations, evaluate $f(2)$.



a. 17

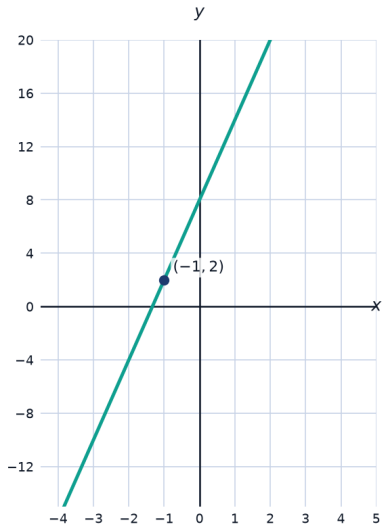
b. 22

c. 10

d. 21

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 8$ in solutions to equations, evaluate $f(2)$.



a. 20

b. 26

c. 12

d. 24

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for introduction to equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for solutions to equations consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The diagram shows point A, point B, and their images A' and B' after a transformation. The transformation can be described by the vector $(p; q)$. Find the value of p.

a. 3

b. -3

c. 5

d. -5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Point $P(-2, 4)$ is translated by the vector $(2; 1)$. Find the image of P.

a. $(0, 5)$

b. $(-4, 3)$

c. $(0, 4)$

d. $(-2, 5)$

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Rotate the point $(1, 7)$ by 180° about the origin. Find the image.

a. $(1, -7)$

b. $(-1, -7)$

c. $(1, 7)$

d. $(-1, 7)$

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)
Rotate the point $(3, -5)$ by 270° anticlockwise about the origin. Find the image.

a. $(-5, -3)$

b. $(3, 5)$

c. $(-3, -5)$

d. $(3, -5)$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Rotate the point $(4, 5)$ by 90° clockwise about the origin. Find the image.

a. $(-4, 5)$

b. $(5, -4)$

c. $(4, 5)$

d. $(4, -5)$

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
Reflect the point $(-1, 2)$ in the line $y = x$. Find the image.

a. $(2, 1)$

b. $(-1, 2)$

c. $(1, -2)$

d. $(2, -1)$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Reflect the point $(9, 1)$ in the x-axis. Find the image.

a. $(1, -9)$

b. $(-9, -1)$

c. $(9, 1)$

d. $(9, -1)$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)
The graph of $y = f(x)$ is shown. Describe the transformations to get $y = f(x) + 2$ and $y = f(-x)$.

a. Translation up 2; reflection in y-axis

b. Translation up 2; reflection in x-axis

c. Translation right 2; reflection in y-axis

d. No transformation

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the magnitude of the vector $\langle 5; 12 \rangle$.

a. 13

b. 24

c. 22

d. 16

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Point $(6, 5)$ is enlarged by scale factor 2 about the origin. Find the image.

a. $(10, 8)$

b. $(14, 10)$

c. $(12, 10)$

d. $(12, 12)$

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Point $(6, 5)$ is enlarged by scale factor 3 about $(3, 2)$. Find the image.

a. $(10, 9)$

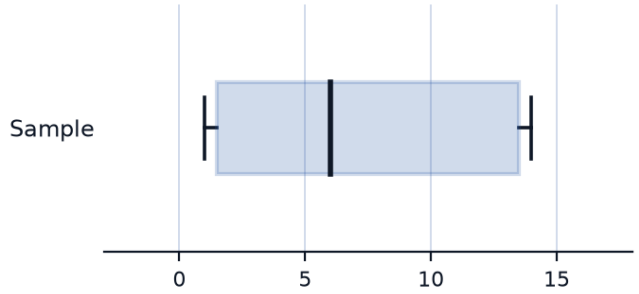
b. $(12, 11)$

c. $(12, 13)$

d. $(14, 11)$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the median of the data set: $1, 2, 6, 13, 14$.



a. 9

b. 13

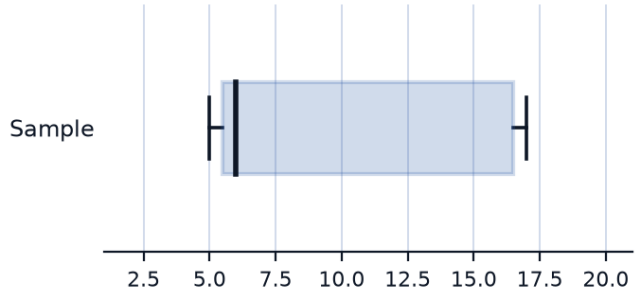
c. 6

d. 7

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the mean of the data set: $5, 6, 6, 16, 17$.



a. 10

b. 14

c. 9

d. 0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

The function f is defined by $f(x) = 2x^2 - 3x + 1$. Find the value of $f(-3)$.

a. 28

b. 10

c. -8

d. -26

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Probability & Statistics

The probability that it rains on Monday is 0.3. If it rains on Monday, the probability it rains on Tuesday is 0.6. If it does not rain on Monday, the probability it rains on Tuesday is 0.2. What is the probability that it rains on at least one of the two days?

a. 0.44

b. 0.56

c. 0.50

d. 0.62

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 9.01 Functions

On a histogram, what does the area of each bar represent?

a. The class width

b. The frequency

c. The frequency density

d. The cumulative frequency

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number

Convert the recurring decimal $0.\dot{2}\dot{7}$ into a fraction in its simplest form.

a. $\frac{27}{100}$

b. $\frac{3}{11}$

c. $\frac{9}{33}$

d. $\frac{1}{4}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Sets & Venn Diagrams

The length of a rectangle is 12 cm correct to the nearest millimetre, and its width is 8 cm correct to the nearest millimetre. Calculate the upper bound for the area of the rectangle in cm^2.

a. (97.0025)

b. (96.0000)

c. (95.0025)

d. (98.0125)

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Algebra & Graphs

A cumulative frequency graph for 120 marathon runners shows that the 90th percentile finish time is 3 hours 45 minutes. How many runners finished in a time faster than 3 hours 45 minutes?

a. 12 runners

b. 90 runners

c. 108 runners

d. 100 runners

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Coordinate Geometry

Rationalise the denominator of $\frac{6}{\sqrt{3}}$.

a. $2\sqrt{3}$

b. $3\sqrt{3}$

c. $\sqrt{2}$

d. $6\sqrt{3}$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Mensuration

Work out $\frac{3}{5} \div \frac{9}{10}$. Give your answer as a fraction in its simplest form.

a. $\frac{2}{3}$

b. $\frac{27}{50}$

c. $\frac{3}{2}$

d. $\frac{1}{3}$

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